LINUX DYNAMIC MEMORY ALLOCATION AND MEMORY MONITORING SYSTEM

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PROJECT OVERVIEW

PROJECT FOCUS

- Linux Dynamic Memory Allocation and Memory Monitoring System
- A Linux-based system that manages, tracks, and monitors dynamic memory allocation and deallocation during program execution.

PROJECT SCOPE

- Dynamic memory allocation and deallocation
- First Fit, Best Fit and Worst Fit
- Memory monitoring using Linux system interfaces
- Fragmentation and leak analysis
- Performance evaluation using different workloads

PROJECT PURPOSE

- Understand dynamic memory allocation in Linux.
- Monitor memory usage and active memory blocks.
- Analyze fragmentation and possible memory leaks.
- Compare different memory allocation strategies.

Problem Statement

1

MEMORY MANAGEMENT CHALLENGE

Frequent memory allocation and deallocation can lead to inefficient memory utilization, fragmentation, and memory leaks.

2

MONITORING CHALLENGE

Memory allocation and usage are difficult to observe and analyze directly during program execution.

3

NEED FOR COMPARISON

Different allocation strategies can produce different memory-utilization and fragmentation patterns, creating a need for systematic performance evaluation.

PROJECT OBJECTIVES

MEMORY MANAGEMENT

Implement dynamic memory allocation, deallocation, and reallocation in a Linux environment using memory-management functions such as malloc(), free(), and realloc(). Maintain records of allocated memory blocks and their status during execution.

MEMORY MONITORING

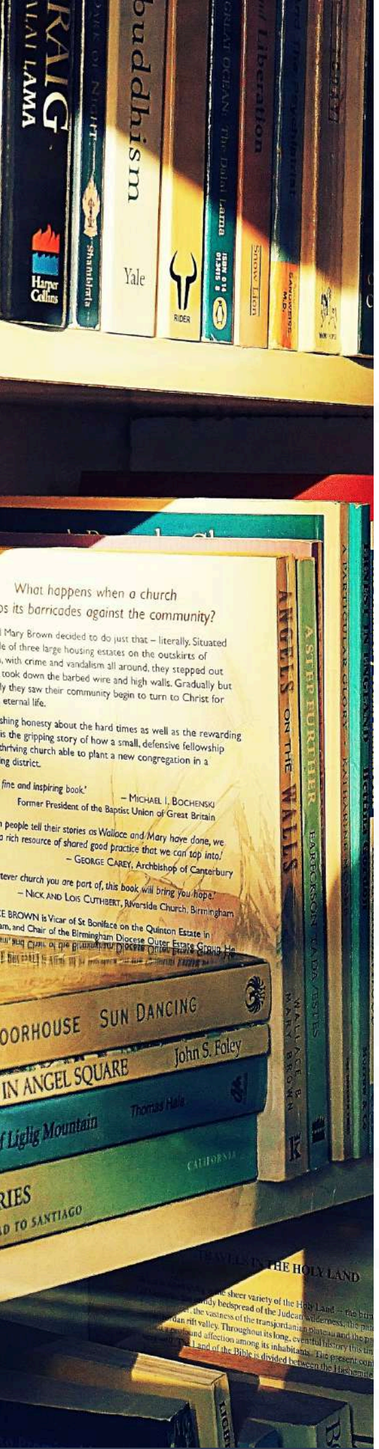
Monitor memory usage and track allocation and deallocation activities during program execution. Use Linux system-level information to observe active memory blocks, memory utilization, peak usage, and identify possible memory leaks.

PERFORMANCE EVALUATION

Test First Fit, Best Fit, and Worst Fit using different memory-allocation workloads. Compare their memory utilization, allocation efficiency, fragmentation, and selected performance metrics to determine their behavior under different conditions.

PROJECT ABSTRACT

- Linux-based system for dynamic memory allocation and deallocation.
- Tracks active memory blocks and allocation/deallocation operations during execution.
- Monitors memory usage and provides memory-related statistics.
- Analyzes memory fragmentation and identifies possible memory leaks.
- Implements First Fit, Best Fit, and Worst Fit allocation strategies.
- Tests the strategies using different memory-allocation workloads.
- Compares memory utilization, allocation efficiency, and selected performance metrics.
- Generates analysis and performance reports based on the collected results.



PROPOSED WORKFLOW

1. MEMORY REQUEST

Program requests a block of memory (size).

2. ALLOCATION STRATEGY

Select an allocation strategy:

- First Fit
- Best Fit
- Worst Fit

3. MEMORY ALLOCATION

Allocate the memory block using the chosen strategy.

4. MEMORY MONITORING

Monitor the memord:

- Active blocks
- Memory usage
- Allocation details

5. DEALLOCATION

Release the memory block when no longer needed.

6. ANALYSIS

Analyze system for:

- Fragmentation
- Memory leaks
- Utilization

SYSTEM MONITORING (Continuous)

- Memory Utilization
- Active Memory Blocks
- Allocation/ Deallocation Count
- Leak Detection

7. REPORT GENERATION

Generate statistics and comparison reports for all strategies.

Next Request

Allocation Strategies



First Fit: allocates the first block large enough to satisfy the request.



Best Fit: allocates the smallest block that still satisfies the request.



Worst Fit: allocates the largest available block for the request.
Compared based on memory utilization and allocation efficiency.

Technologies / APIs

Platform & Tools:
Linux/Ubuntu, C, GCC, GDB, Git & GitHub

Linux Memory APIs:
malloc() (allocate), free() (release), realloc() (resize), mmap()/munmap() (memory mapping), /proc (process & memory info)

EXPECTED OUTCOMES & CONCLUSION

1. Functional System
A reliable Linux-based system for dynamic memory allocation, deallocation, and monitoring.

2. Real-time Monitoring
Real-time tracking of memory allocation and deallocation activities during program execution.

3. Memory Analysis
Detailed analysis of memory utilization, fragmentation, and memory behavior.

4. Leak Detection
Identification of possible memory leaks and abnormal memory usage patterns.

5. Strategy Comparison
Performance comparison of First Fit, Best Fit, and Worst Fit allocation strategies under different workloads.

6. Reports & Insights
Generation of memory statistics and comparison reports for better insights and decision making.

The project will provide a better understanding of dynamic memory management in Linux and help in choosing the most efficient allocation strategy for optimal memory utilization.

THANK YOU